Brady Bangasser

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Compiler-focused engineer working close to the metal: LLVM IR optimization passes, toolchain performance, and language design, applied to real high-performance workloads.

EXPERIENCE

Research Assistant — Compiler Optimization, Iowa State University Aug 2025 – Present

- Built OMPRTF (OpenMP RunTime Fabric) for Iowa State's SWAPP lab, a custom OpenMP runtime and dynamic-analysis pipeline that applies aggressive, conditional optimizations to GPU-offload programs by lowering them to LLVM IR.
- Wrote an LLVM optimization pass that breaks opaque OpenMP runtime calls into discrete IR instructions, pruning redundant host-GPU data movement without destabilizing kernels.

PROJECTS

Omni HTTP Router 2025–2026

- Built a Rust HTTP router that resolves middleware at compile time into fully static stacks, routing a request in under 30 instructions on SIMD-enabled x86.
- Wrote a compiler that lowers a routing DSL to native object files and generates matching Terraform and binaries for easy, fast cloud deployment.

Minnesotan Programming Language 2023–2025

- Designed a Turing-complete esoteric language and built an end-to-end compiler with Flex/Bison that emits LLVM IR, compiling to native executables.
- Implemented a custom lexer and parser that transform the language's colloquial syntax into an AST before LLVM IR emission.
- Wrote the grammar and documentation so others can author and compile Minnesotan programs.

Public Data Scraping and Processing 2025

- Wrote a compiler-like tool that pairs lexical analysis with an LLM to standardize and validate data formats, and trained an ML model to flag anomalies in the collected data.
- Built a parallel, scalable pipeline to scrape, clean, and store public datasets, deployed via a Terraform CI/CD pipeline to AWS and a personal Kubernetes cluster.

EDUCATION

M.S. Computer Science, Iowa State University Aug 2025 – May 2027

GPA: 4.00

- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

B.S. Computer Science, Iowa State University Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, CUDA, Lua
Compilers & HPC: LLVM, Flex/Bison, OpenMP